

Package ‘complexbrowser’

June 23, 2026

Type Package

Title Protein Complex Analysis Workflows

Version 0.0.0.9000

Author ComplexBrowser contributors

Maintainer ComplexBrowser contributors <veits@bmb.sdu.dk>

Description Reusable workflows for ComplexBrowser proteomics quality control and protein complex analysis, with support for the existing Shiny application, R package APIs, and non-interactive command-line integration.

License GPL-3

Encoding UTF-8

Depends R (>= 3.5.0)

Imports gtools, limma, preprocessCore, qvalue

Suggests knitr, rmarkdown, shiny, testthat (>= 3.0.0)

VignetteBuilder knitr

Config/testthat/edition 3

RoxygenNote 7.3.3

Contents

complexbrowser-package	2
complexbrowser_calculate_statistics	3
complexbrowser_changing_table_data	4
complexbrowser_cli	5
complexbrowser_complex_summary	5
complexbrowser_complex_table_filename	6
complexbrowser_display_complex_table	6
complexbrowser_extdata_path	7
complexbrowser_filter_database	7
complexbrowser_filter_stats_for_database	8
complexbrowser_input_statistics_filename	8
complexbrowser_load_complex_portal	9

complexbrowser_load_corum	9
complexbrowser_prepare_input_table	10
complexbrowser_prepare_qc_report_data	10
complexbrowser_prepare_user_database	11
complexbrowser_run_app	11
complexbrowser_run_complex_analysis	12
complexbrowser_run_qc	13
complexbrowser_score_complexes	14
complexbrowser_select_database	14
complexbrowser_write_complex_table	15
complexbrowser_write_input_statistics	15
Index	16

complexbrowser-package

ComplexBrowser proteomics quality-control and protein-complex workflows

Description

The **complexbrowser** package exposes reusable R functions for the ComplexBrowser Shiny application. It supports tabular proteomics input preparation, QC/statistics calculation, bundled complex database loading, protein-complex matching and scoring, CSV/RDS exports, vignettes, and a thin command-line interface for headless workflows.

Details

Main workflow functions:

- `complexbrowser_run_qc()` prepares input data and calculates QC statistics.
- `complexbrowser_run_complex_analysis()` matches quantified proteins to CORUM, EBI Complex Portal, or user-defined complex databases.
- `complexbrowser_cli()` runs the package workflows from command-line-style arguments.
- `complexbrowser_run_app()` starts the packaged Shiny application.

Bundled data:

- Prepared CORUM and EBI Complex Portal databases are available through `complexbrowser_load_corum()` and `complexbrowser_load_complex_portal()`.
- Reusable example inputs are available below `system.file("extdata", "examples", package = "complexbrowser")`.
- Shiny-specific example inputs are available below `system.file("shiny", "data", package = "complexbrowser")`.

See Also

[complexbrowser_run_qc](#), [complexbrowser_run_complex_analysis](#), [complexbrowser_cli](#), [complexbrowser_run_app](#),
`vignette("qc-workflow", package = "complexbrowser")`, `vignette("complex-analysis", package = "complexbrowser")`

Examples

```
library(complexbrowser)

input_file <- system.file(
  "shiny", "data", "example_tcell_cut.csv",
  package = "complexbrowser"
)
input <- read.csv(input_file, check.names = FALSE)

qc <- suppressWarnings(complexbrowser_run_qc(
  input,
  no_cond = 4,
  no_rep = 2,
  log2 = FALSE,
  grouped = TRUE,
  q_values = FALSE
))

qc$metadata

sort(unique(complexbrowser_load_corum()$Organism))
```

`complexbrowser_calculate_statistics`*Calculate ComplexBrowser QC statistics*

Description

Calculate ComplexBrowser QC statistics

Usage

```
complexbrowser_calculate_statistics(
  data,
  no_cond,
  no_rep,
  q_values = TRUE,
  normalization = NULL,
  design = c("unpaired", "paired")
)
```

Arguments

data	Prepared input table with protein identifiers in the first column.
no_cond	Number of experimental conditions.
no_rep	Number of replicates per condition.
q_values	If 'TRUE', q-values are read from 'data'; otherwise they are calculated with LIMMA.
normalization	Optional normalization method: '"Total Intensity"', '"Mean"', '"Median"', or '"Quantile"'.
design	Statistical design used when q-values must be calculated: '"unpaired"' or '"paired"'.

Value

A named list of statistics matching the legacy Shiny output shape.

complexbrowser_changing_table_data
Build top changing-complex table data

Description

Build top changing-complex table data

Usage

```
complexbrowser_changing_table_data(
  scored_database,
  condition,
  noise_threshold,
  fc_threshold
)
```

Arguments

scored_database	Scored complex database.
condition	Condition number.
noise_threshold	Maximum allowed noise.
fc_threshold	Fold-change threshold.

Value

A data frame for top up/down complexes.

complexbrowser_cli	<i>Run the ComplexBrowser command-line interface</i>
--------------------	--

Description

Run the ComplexBrowser command-line interface

Usage

```
complexbrowser_cli(args = commandArgs(trailingOnly = TRUE))
```

Arguments

args	Command-line arguments. Defaults to 'commandArgs(TRUE)'.
------	--

Value

An integer status code, invisibly: 0 for success, 2 for usage errors, and 1 for unexpected runtime errors.

complexbrowser_complex_summary	<i>Summarize scored complexes for a condition</i>
--------------------------------	---

Description

Summarize scored complexes for a condition

Usage

```
complexbrowser_complex_summary(  
  scored_database,  
  no_cond,  
  no_rep,  
  condition,  
  noise_threshold  
)
```

Arguments

scored_database	Scored complex database.
no_cond	Number of experimental conditions.
no_rep	Number of replicates per condition.
condition	Condition number.
noise_threshold	Maximum allowed noise.

Value

A character scalar summary.

complexbrowser_complex_table_filename
<i>Build the complex-table CSV filename</i>

Description

Build the complex-table CSV filename

Usage

```
complexbrowser_complex_table_filename(time = Sys.time())
```

Arguments

time Timestamp to include in the filename.

Value

A filename matching the Shiny download handler.

complexbrowser_display_complex_table
<i>Convert scored complexes to the display-table shape</i>

Description

Convert scored complexes to the display-table shape

Usage

```
complexbrowser_display_complex_table(scored_database)
```

Arguments

scored_database Filtered database with appended score columns.

Value

A data frame with subunits collapsed for display.

`complexbrowser_extdata_path`*Locate a packaged ComplexBrowser extdata file*

Description

Locate a packaged ComplexBrowser extdata file

Usage

```
complexbrowser_extdata_path(..., must_work = TRUE)
```

Arguments

<code>...</code>	Path components below the package ‘inst/extdata’ directory.
<code>must_work</code>	If ‘TRUE’, error when the resolved file does not exist.

Value

A character scalar with the resolved path.

`complexbrowser_filter_database`*Filter a complex database to complexes with quantified subunits*

Description

Filter a complex database to complexes with quantified subunits

Usage

```
complexbrowser_filter_database(f_data, database, organism)
```

Arguments

<code>f_data</code>	Filtered absolute intensity data.
<code>database</code>	Prepared complex database.
<code>organism</code>	Organism label to analyze.

Value

A filtered database, or ‘NULL’ when no complexes match.

`complexbrowser_filter_stats_for_database`*Filter statistics to proteins present in a complex database*

Description

Filter statistics to proteins present in a complex database

Usage

```
complexbrowser_filter_stats_for_database(stats, database, organism)
```

Arguments

<code>stats</code>	QC statistics list.
<code>database</code>	Prepared complex database or database selector accepted by ‘complexbrowser_select_database()’.
<code>organism</code>	Organism label to analyze.

Value

A statistics list filtered to proteins found in database complexes.

`complexbrowser_input_statistics_filename`*Build the input-statistics CSV filename*

Description

Build the input-statistics CSV filename

Usage

```
complexbrowser_input_statistics_filename(time = Sys.time())
```

Arguments

<code>time</code>	Timestamp to include in the filename.
-------------------	---------------------------------------

Value

A filename matching the Shiny download handler.

`complexbrowser_load_complex_portal`*Load the bundled prepared EBI Complex Portal database*

Description

Load the bundled prepared EBI Complex Portal database

Usage

```
complexbrowser_load_complex_portal()
```

Value

A data frame containing prepared EBI Complex Portal complex records.

`complexbrowser_load_corum`*Load the bundled prepared CORUM database*

Description

Load the bundled prepared CORUM database

Usage

```
complexbrowser_load_corum()
```

Value

A data frame containing prepared CORUM complex records.

`complexbrowser_prepare_input_table`*Prepare a quantitative proteomics input table*

Description

Prepare a quantitative proteomics input table

Usage

```
complexbrowser_prepare_input_table(  
  data,  
  no_cond,  
  no_rep,  
  log2 = TRUE,  
  grouped = FALSE,  
  q_values = TRUE  
)
```

Arguments

<code>data</code>	A data frame whose first column contains protein identifiers.
<code>no_cond</code>	Number of experimental conditions.
<code>no_rep</code>	Number of replicates per condition.
<code>log2</code>	If 'TRUE', measurement values are treated as log2 values and converted to absolute intensities.
<code>grouped</code>	If 'TRUE', replicate columns are grouped by condition.
<code>q_values</code>	If 'TRUE', q-value columns are expected after measurement columns.

Value

A data frame with ComplexBrowser-compatible column names and order.

`complexbrowser_prepare_qc_report_data`*Prepare QC report summary data*

Description

Prepare QC report summary data

Usage

```
complexbrowser_prepare_qc_report_data(stats, no_cond, no_rep)
```

Arguments

stats	QC statistics list.
no_cond	Number of experimental conditions.
no_rep	Number of replicates per condition.

Value

A list matching the legacy ‘QCreport.rmd’ parameter data.

`complexbrowser_prepare_user_database`

Prepare a user-defined complex database

Description

Prepare a user-defined complex database

Usage

```
complexbrowser_prepare_user_database(csv_file)
```

Arguments

csv_file	Path to a CSV file with complex database columns.
----------	---

Value

A prepared database data frame.

`complexbrowser_run_app`

Launch the packaged ComplexBrowser Shiny app

Description

Start the installed ComplexBrowser app from the package ‘inst/shiny’ directory.

Usage

```
complexbrowser_run_app(...)
```

Arguments

...	Additional arguments passed to ‘shiny::runApp()’.
-----	---

Value

The result of 'shiny::runApp()', invisibly.

Examples

```
## Not run:  
complexbrowser_run_app()  
  
## End(Not run)
```

```
complexbrowser_run_complex_analysis  
  Run package-level complex analysis
```

Description

Run package-level complex analysis

Usage

```
complexbrowser_run_complex_analysis(  
  stats,  
  database,  
  organism,  
  no_cond,  
  no_rep,  
  database_name = NULL  
)
```

Arguments

stats	QC statistics list.
database	Prepared complex database.
organism	Organism label to analyze.
no_cond	Number of experimental conditions.
no_rep	Number of replicates per condition.
database_name	Optional database label for metadata.

Value

A 'complexbrowser_complex_result' object.

complexbrowser_run_qc *Run the package QC workflow*

Description

Run the package QC workflow

Usage

```
complexbrowser_run_qc(  
  data,  
  no_cond,  
  no_rep,  
  log2 = TRUE,  
  grouped = FALSE,  
  q_values = TRUE,  
  normalization = NULL,  
  design = c("unpaired", "paired")  
)
```

Arguments

data	A data frame whose first column contains protein identifiers.
no_cond	Number of experimental conditions.
no_rep	Number of replicates per condition.
log2	If 'TRUE', measurement values are treated as log2 values and converted to absolute intensities.
grouped	If 'TRUE', replicate columns are grouped by condition.
q_values	If 'TRUE', q-value columns are expected after measurement columns.
normalization	Optional normalization method: "Total Intensity", "Mean", "Median", or "Quantile".
design	Statistical design used when q-values must be calculated: "unpaired" or "paired".

Value

A 'complexbrowser_qc_result' object with prepared input, statistics, merged display table, and workflow metadata.

complexbrowser_score_complexes
<i>Score filtered complexes with the legacy FARMS workflow</i>

Description

Score filtered complexes with the legacy FARMS workflow

Usage

complexbrowser_score_complexes(filtered_database, stats, no_cond, no_rep)

Arguments

- filtered_database
Database returned by 'complexbrowser_filter_database()'.
- stats
Filtered statistics returned by 'complexbrowser_filter_stats_for_database()'.
- no_cond
Number of experimental conditions.
- no_rep
Number of replicates per condition.

Value

A matrix of complex fold-change and noise scores.

complexbrowser_select_database
<i>Select a bundled or supplied complex database</i>

Description

Select a bundled or supplied complex database

Usage

complexbrowser_select_database(database)

Arguments

- database
Database selector. Use "CORUM", "EBI Complex Portal", a prepared data frame, or a path to a user-defined CSV database.

Value

A prepared complex database.

`complexbrowser_write_complex_table`*Write the complex-analysis table as CSV*

Description

Write the complex-analysis table as CSV

Usage

```
complexbrowser_write_complex_table(table, file)
```

Arguments

<code>table</code>	Complex analysis display table.
<code>file</code>	Output CSV path.

Value

'file', invisibly.

`complexbrowser_write_input_statistics`*Write the input-statistics table as CSV*

Description

Write the input-statistics table as CSV

Usage

```
complexbrowser_write_input_statistics(table, file)
```

Arguments

<code>table</code>	Input statistics table.
<code>file</code>	Output CSV path.

Value

'file', invisibly.

Index

complexbrowser
 (complexbrowser-package), [2](#)
complexbrowser-package, [2](#)
complexbrowser_calculate_statistics, [3](#)
complexbrowser_changing_table_data, [4](#)
complexbrowser_cli, [2](#), [5](#)
complexbrowser_complex_summary, [5](#)
complexbrowser_complex_table_filename,
 [6](#)
complexbrowser_display_complex_table,
 [6](#)
complexbrowser_extdata_path, [7](#)
complexbrowser_filter_database, [7](#)
complexbrowser_filter_stats_for_database,
 [8](#)
complexbrowser_input_statistics_filename,
 [8](#)
complexbrowser_load_complex_portal, [9](#)
complexbrowser_load_corum, [9](#)
complexbrowser_prepare_input_table, [10](#)
complexbrowser_prepare_qc_report_data,
 [10](#)
complexbrowser_prepare_user_database,
 [11](#)
complexbrowser_run_app, [2](#), [11](#)
complexbrowser_run_complex_analysis, [2](#),
 [12](#)
complexbrowser_run_qc, [2](#), [13](#)
complexbrowser_score_complexes, [14](#)
complexbrowser_select_database, [14](#)
complexbrowser_write_complex_table, [15](#)
complexbrowser_write_input_statistics,
 [15](#)